eBGP, iBGP, EIGRP and OSPF configuration

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Purpose:

The purpose of this lab is to learn about the configuration of iBGP (interior Border Gateway Protocol) which has become a cornerstone of modern networking, due to the flexibility associated with route sharing not available across IGPs (Interior Gateway Protocols) such as metric retention, lower overhead, and an increased route-sharing bandwidth

Background Information:

Border Gateway Protocol (BGP) is an essential routing protocol used for transferring data across the internet, specifically between large networks, known as autonomous systems (ASs), such as major enterprise networks. BGP plays a critical role in managing the complex paths that data takes as it travels across these networks, ensuring that data packets reach their destination efficiently. It has two main modes: internal BGP (iBGP) for communication within a single AS, and external BGP (eBGP) for routing data between different ASs. BGP relies on TCP (Transmission Control Protocol) over port 179, which makes it reliable because TCP itself ensures data packets are properly delivered and in the right order. One of the primary features of eBGP is its path-vector mechanism, which maintains a record of AS numbers, allowing it to avoid routing loops by tracking where data has been. This feature also makes it highly scalable for complex networks, as it can handle vast amounts of routing information. eBGP is known for its policy-based routing, which allows network administrators to set up rules that dictate data flow based on factors beyond simple distance, such as security policies or business agreements. This scalability and control make eBGP essential to the internet’s infrastructure.

iBGP (interior Border Gateway Protocol) handles large routing tables more efficiently than IGPs (Interior Gateway Protocols) like OSPF (Open Shortest Path First) or EIGRP (Enhanced Interior Gateway Routing Protocol) because it focuses on route policies rather than calculating the entire network's shortest paths. IGPs require each router to store and process a complete view of the network topology,

which becomes inefficient as the number of routes grows. Instead, iBGP simplifies this by only storing the best paths to destinations, using attributes like AS-path and next-hop, which reduces the processing burden on routers. Additionally, iBGP provides more control over traffic paths by allowing administrators to influence routing decisions using various attributes. For example, AS-path length can be adjusted to prioritize shorter paths, local preference can be set to make certain routes more favorable, and the Multi-Exit Discriminator (MED) can help manage inbound traffic. This flexibility allows for better traffic management and load balancing compared to IGPs, which rely on automatic metrics like link cost. Furthermore, iBGP reduces the risk of routing loops, a common issue in complex networks. It uses techniques like the split horizon rule, which prevents routes learned from one iBGP peer from being advertised back to that peer. The standard full mesh requirement also ensures that all iBGP peers are directly connected, giving each router a clear view of route information without loops. When full mesh is impractical, solutions like Route Reflectors or Confederations are used to maintain loop-free routing. Additionally, using loopback interfaces for peering adds stability, as these virtual addresses remain consistent even if physical interfaces fail. These features make iBGP a more scalable and stable choice for large networks, offering better performance and more precise control over routing than traditional IGPs.

Enhanced Interior Gateway Routing Protocol (EIGRP) is a Cisco-proprietary protocol, meaning it was originally designed only for Cisco devices, although it's now sometimes used more widely. EIGRP combines elements from both distance-vector and link-state protocols, and it's designed specifically for routing within a single AS. EIGRP is widely used in large enterprise networks because of its speed and efficiency. Its Diffusing Update Algorithm (DUAL) ensures that there are loop-free paths and that the network converges quickly, which means it can adapt to changes, like new routes or failed links, in real-time without causing delays. Instead of sending out entire routing tables like some other protocols, EIGRP only sends updates when there's a change, which conserves bandwidth and reduces unnecessary network traffic. EIGRP also uses a unique composite metric, which takes into account factors like bandwidth, delay, load, and reliability, rather than just distance. This makes it easier for network administrators to find the most efficient path for data while also managing the reliability and load of each route.

Open Shortest Path First (OSPF) is an open-standard, link-state protocol that's commonly used for routing within ASs, especially in multi-vendor environments, meaning networks that don't rely on just one manufacturer's

devices. OSPF creates a complete map of the network using a synchronized link-state database (LSDB) that all OSPF routers share by exchanging link-state advertisements (LSAs). This topological map of the network helps each router have a full view of the entire network structure, allowing them to calculate optimal routes based on the entire network's layout. OSPF is also highly organized and scalable, thanks to its hierarchical design. It uses areas to group routers and links, making it possible to reduce the load and traffic within large networks by focusing updates in specific parts of the network rather than everywhere at once, however we are using only one area of OSPF. For determining the shortest path, OSPF relies on Dijkstra's Shortest Path First (SPF) algorithm, which finds the least-cost route to each destination. The cost is typically determined by bandwidth, but it can be adjusted to consider other factors if needed.

Summary:

In this lab I configure 7 routers, 2 separate EIGRP instances, 2 separate eBGP instances, 1 iBGP instance and 1 OSPF instance, this is meant to model the connectivity between and within corporate networks, being connected by the BGP instances for interconnectivity between the networks. The BGP took and gave routes to both the EIGRP and OSPF instances to allow end to end connectivity. And iBGP worked to optimize the route sharing between autonomous systems 2 and 5

Commands:

Routers 1 & 6

hostname R1 – assigns hostname to router
ipv6 unicast-routing – allows ipv6 to be routed
interface GigabitEthernet0/0/0 – allows you to configure the interface
ip address 192.168.1.1 255.255.255.252 – assigns ip address based on address table
ipv6 address FE80::1 link-local – assigns ipv6 link-local address based on address table
ipv6 address 2001:DB8:ACAD:B::1/64 – assigns ipv6 address based on address table
ipv6 enable – allows the interface to run ipv6 routing protocols
ipv6 eigrp 10 – configures the interface with ipv6 eigrp
interface GigabitEthernet0/0/1 – allows you to configure the interface
ip address 192.168.0.2 255.255.255.0 – assigns ip address based on address table
ipv6 address FE80::2 link-local – assigns ipv6 link-local address based on address table
ipv6 address 2001:DB8:ACAD:A::2/64 – assigns interface ipv6 address based on address table

ipv6 enable – enables interface to run ipv6 routing
ipv6 eigrp 10 – configures ipv6 eigrp on the interface
router eigrp 10 – allows configuration of the eigrp routing protocol
network 192.168.0.0 0.0.0.3 – allows the network stated to run eigrp through the router
network 192.168.1.0 0.0.0.3 – allows the network stated to run eigrp through the router
eigrp router-id 1.1.1.1 – assigns the eigrp router-id
ipv6 router eigrp 10 – allows configuration of the eigrp routing protocol
eigrp router-id 1.1.1.1 – assigns the ipv6 router-id

Routers 2 & 5

hostname R2 – assigns hostname to router
ipv6 unicast-routing – allows ipv6 to be routed
interface GigabitEthernet0/0/0 – allows you to configure the interface
ip address 192.168.2.1 255.255.255.252 – assigns ip address based on address table
ipv6 address FE80::2 link-local – assigns ipv6 link-local address based on address table
ipv6 address 2001:DB8:ACAD:C::1/64 – assigns ipv6 address based on address table
ipv6 enable – enables ipv6 routing protocols on the interface
ipv6 eigrp 10 – sets up ipv6 eigrp on this interface
interface GigabitEthernet0/0/1 – allows you to configure the interface
ip address 192.168.1.2 255.255.255.252 – assigns ip address based on address table
ipv6 address FE80::2 link-local – assigns ipv6 link-local address based on address table
ipv6 address 2001:DB8:ACAD:B::2/64 – assigns ipv6 address based on address table
ipv6 enable – enables ipv6 routing protocols on the interface
ipv6 eigrp 10 – configures ipv6 eigrp to run on the interface
router eigrp 10 – allows configuration of eigrp with autonomous system number 10
network 192.168.1.0 0.0.0.3 – allows network stated to run eigrp through the router
network 192.168.2.0 0.0.0.3 – allows network stated to run eigrp through the router

redistribute connected metric 10 0 255 255 1 – redistributes the networks on router to eBGP
redistribute bgp 2 metric 10 0 255 255 1 – redistributes eBGP networks back to eigrp network
eigrp router-id 2.2.2.2 – sets router id for eigrp
router bgp 2 – allows configuration of bgp
bgp router-id 2.2.2.2 – sets the bgp router id
bgp log-neighbor-changes – sends notification and logs when a neighbor adjacency is set up or broken
neighbor 2001:DB8:ACAD:C::2 remote-as 3 – declares set ip address as the neighbor
neighbor 192.168.2.2 remote-as 3 – sets up to look for the ip address of neighbor with matching ip's
address-family ipv4 – allows specific configuration of ipv4 bgp
redistribute connected metric 10 – sets connects redistribution of AS of the router and BGP
redistribute eigrp 10 metric 10 – redistributes the eigrp network to BGP
no neighbor 2001:DB8:ACAD:C::2 activate – makes sure ipv6 neighbor doesn't overlap ipv4
neighbor 192.168.2.2 activate – allows ipv4 neighbor to share and shares routes with that neighbor
exit-address-family – exits address family
address-family ipv6 – allows configuration of ipv6 bgp specifically
redistribute connected metric 10 – connects redistribution of AS and bgp
redistribute eigrp 10 metric 10 – redistributes the eigrp networks to bgp
neighbor 2001:DB8:ACAD:C::2 activate – allows the ipv6 neighbor to add routes
exit-address-family – exits address family
ipv6 router eigrp 10 – allows configuration of ipv6 eigrp
eigrp router-id 2.2.2.2 – sets the router id of ipv6 eigrp
redistribute bgp 2 metric 10 0 255 255 1 – allows redistribution of bgp routes to the eigrp networks

Routers 3 & 4

hostname R3 – assigns hostname to router
ipv6 unicast-routing – allows ipv6 to be routed
interface Loopback0 – allows you to configure loopback interface, an interface connected to nothing
ip address 10.0.0.1 255.255.255.255 – sets a loopback ip address
ipv6 address FE80::1 link-local – sets a ipv6 link local address
ipv6 address 2001:DB8:ACAD:1::1/128 – sets an ipv6 loopback address

ipv6 enable – enables ipv6 configurations on interface
ipv6 ospf 10 area 0 – sets ipv6 ospf area and AS number
interface GigabitEthernet0/0/0 – allows you to configure the interface
ip address 192.168.3.1 255.255.255.252 – assigns ip address based on address table
ip ospf 10 area 0 – configures the interface with ospf and the area
ipv6 address FE80::1 link-local – assigns ipv6 link-local address based on address table
ipv6 address 2001:DB8:ACAD:D::1/64 – assigns ipv6 address based on address table
ipv6 enable – enable ipv6 configuration on interface
ipv6 ospf 10 area 0 – configures ipv6 ospf and area
interface GigabitEthernet0/0/1 – allows you to configure the interface
ip address 192.168.2.2 255.255.255.252 – configures ip address based on address table
ipv6 address FE80::1 link-local – configures ipv6 link-local address based on address table
ipv6 address 2001:DB8:ACAD:C::2/64 – configures ipv6 address based on address table
router ospf 10 – allows configuration of ospf
router-id 3.3.3.3 – configures opsf router id
redistribute bgp 3 metric 10 subnets – redistributes the bgp routes to ospf
network 10.0.0.1 0.0.0.0 area 0 – sets up ospf network to loopback interface
network 192.168.2.0 0.0.0.3 area 0 – sets up ospf networks to neighboring routers
network 192.168.3.0 0.0.0.3 area 0 – sets up ospf networks to neighboring routers
router bgp 3 – allows configuration to bgp
bgp router-id 3.3.3.3 – configures bgp router id
neighbor 10.0.0.2 remote-as 3 – sets the loopback interface of opposite boarder router as the neighbor
neighbor 10.0.0.2 update-source Loopback0 – allows the loopback interface to update information
neighbor 2001:DB8:ACAD:1::2 remote-as 3 – sets the ipv6 neighbor of the opposite boarder router
neighbor 2001:DB8:ACAD:1::2 update-source Loopback0 – allows the loopback interface to update info
bgp log-neighbor-changes – logs and notifies of neighbor adjacency set ups and break downs
neighbor 2001:DB8:ACAD:C::1 remote-as 2 – sets ipv6 neighbor
neighbor 192.168.2.1 remote-as 2 – sets ipv4 neighbor
address-family ipv4 – allows specific ipv4 configurations to bgp
redistribute connected – connects redistributions from ospf and bgp
redistribute ospf 10 metric 10 – redistributes ospf routes to bgp
neighbor 10.0.0.2 activate – sets partial mesh iBGP connection using loopback interface
no neighbor 2001:DB8:ACAD:1::2 activate – prevents ipv6 neighbor to interfere with ipv4
no neighbor 2001:DB8:ACAD:C::1 activate – makes ipv6 neighbor not interfere with ipv4
neighbor 192.168.2.1 activate – sets ipv4 neighbor communication
exit-address-family – exits address family
address-family ipv6 – allows specific ipv4 configurations to bgp
redistribute connected – connects redistribution from ospf and bgp

redistribute ospf 10 metric 10 – redistributes ospf routes to bgp
neighbor 2001:DB8:ACAD:1::2 activate – activates ipv6 neighbor of opposite boarder router
neighbor 2001:DB8:ACAD:C::1 activate – activates ipv6 neighbor
exit address-family – exits address family
ipv6 router ospf 10 – allows ipv6 configurations for ospf
router-id 3.3.3.3 – sets opsfv3 router id
redistribute bgp 3 – redistributes ipv6 bgp routes

Router 7

hostname R7 – sets hostname
ipv6 unicast-routing – allows ipv6 routing
interface GigabitEthernet0/0/0 – allows interface configuration
ip address 192.168.7.1 255.255.255.252 – sets ip address
ipv6 address FE80::1 link-local – sets ipv6 link local address
ipv6 address 2001:DB8:ACAD:A2::1/64 – sets ipv6 address
ipv6 enable – allows ipv6 configuration on interface
ipv6 ospf 10 area 1 – sets ipv6 opsf area number and AS number
interface GigabitEthernet0/0/1 – allows interface3 configuration
ip address 192.168.3.2 255.255.255.252 – sets ipv4 address
ipv6 address FE80::2 link-local – sets ipv6 link local address
ipv6 address 2001:DB8:ACAD:D::2/64 – sets ipv6 address
ipv6 enable – enables ipv6 configuration on interface
ipv6 ospf 10 area 1 – sets ipv6 ospf AS number and area number
router ospf 10 – allows ipv4 AS config of ospf
router-id 7.7.7.7 – sets router id
network 192.168.3.0 0.0.0.3 area 1 – sets network to neighbor router
network 192.168.7.0 0.0.0.3 area 1 – sets network to neighbor router
ipv6 router ospf 10 – enables ipv6 ospf of AS 10

Configurations:

Router1:

```
Current configuration : 1714 bytes
Last configuration change at 22:46:09 UTC Thu Nov 14 2024
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname R1
boot-start-marker
boot-end-marker
```

```
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FLM2408005M
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 192.168.1.1 255.255.255.252
negotiation auto
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:ACAD:B::1/64
ipv6 enable
ipv6 eigrp 10
interface GigabitEthernet0/0/1
ip address 192.168.0.2 255.255.255.0
negotiation auto
ipv6 address FE80::2 link-local
ipv6 address 2001:DB8:ACAD:A::2/64
ipv6 enable
ipv6 eigrp 10
interface GigabitEthernet0/2/0
no ip address
negotiation auto
interface GigabitEthernet0/2/1
no ip address
negotiation auto
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto
router eigrp 10
network 192.168.0.0
network 192.168.1.0 0.0.0.3
eigrp router-id 1.1.1.1
ip forward-protocol nd
no ip http server
ip http secure-server
```

```
ip tftp source-interface GigabitEthernet0
ipv6 router eigrp 10
eigrp router-id 1.1.1.1
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Router2:

```
Current configuration : 2338 bytes
Last configuration change at 22:49:30 UTC Thu Nov 14 2024
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname R2
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FLM24060912
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 192.168.2.1 255.255.255.252
negotiation auto
ipv6 address FE80::2 link-local
ipv6 address 2001:DB8:ACAD:C::1/64
ipv6 enable
ipv6 eigrp 10
interface GigabitEthernet0/0/1
```

```
ip address 192.168.1.2 255.255.255.252
negotiation auto
ipv6 address FE80::2 link-local
ipv6 address 2001:DB8:ACAD:B::2/64
ipv6 enable
ipv6 eigrp 10
interface GigabitEthernet0/2/0
no ip address
negotiation auto
interface GigabitEthernet0/2/1
no ip address
negotiation auto
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto
router eigrp 10
network 192.168.1.0 0.0.0.3
network 192.168.2.0 0.0.0.3
redistribute connected metric 10 0 255 255 1
redistribute bgp 2 metric 10 0 255 255 1
eigrp router-id 2.2.2.2
router bgp 2
bgp router-id 2.2.2.2
bgp log-neighbor-changes
neighbor 2001:DB8:ACAD:C::2 remote-as 3
neighbor 192.168.2.2 remote-as 3
address-family ipv4
redistribute connected metric 10
redistribute eigrp 10 metric 10
no neighbor 2001:DB8:ACAD:C::2 activate
neighbor 192.168.2.2 activate
exit-address-family
address-family ipv6
redistribute connected metric 10
redistribute eigrp 10 metric 10
neighbor 2001:DB8:ACAD:C::2 activate
exit-address-family
ip forward-protocol nd
no ip http server
ip http secure-server
ipv6 router eigrp 10
eigrp router-id 2.2.2.2
redistribute bgp 2 metric 10 0 255 255 1
redistribute connected metric 10 0 255 255 1
control-plane
line con 0
```

```
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Router3:

```
Current configuration : 2608 bytes
Last configuration change at 22:47:52 UTC Thu Nov 14 2024
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
hostname R3
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
ipv6 unicast-routing
subscriber templating
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FDO214420G7
spanning-tree extend system-id
redundancy
mode none
vlan internal allocation policy ascending
interface Loopback0
ip address 10.0.0.1 255.255.255.255
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:ACAD:1::1/128
ipv6 enable
ipv6 ospf 10 area 0
interface GigabitEthernet0/0/0
ip address 192.168.3.1 255.255.255.252
negotiation auto
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:ACAD:D::1/64
ipv6 enable
ipv6 ospf 10 area 1
interface GigabitEthernet0/0/1
ip address 192.168.2.2 255.255.255.252
negotiation auto
```

```
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:ACAD:C::2/64
ipv6 enable
ipv6 ospf 10 area 1
interface Serial0/1/0
interface Serial0/1/1
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
interface Vlan1
no ip address
shutdown
router ospf 10
router-id 3.3.3.3
redistribute bgp 3 metric 10 subnets
network 10.0.0.1 0.0.0.0 area 0
network 192.168.2.0 0.0.0.3 area 1
network 192.168.3.0 0.0.0.3 area 1
router bgp 3
bgp router-id 3.3.3.3
bgp log-neighbor-changes
neighbor 10.0.0.2 remote-as 3
neighbor 10.0.0.2 update-source Loopback0
neighbor 2001:DB8:ACAD:1::2 remote-as 3
neighbor 2001:DB8:ACAD:1::2 update-source Loopback0
neighbor 2001:DB8:ACAD:C::1 remote-as 2
neighbor 192.168.2.1 remote-as 2
address-family ipv4
redistribute connected
redistribute ospf 10 metric 10
neighbor 10.0.0.2 activate
no neighbor 2001:DB8:ACAD:1::2 activate
no neighbor 2001:DB8:ACAD:C::1 activate
neighbor 192.168.2.1 activate
exit-address-family
address-family ipv6
redistribute connected
redistribute ospf 10 metric 10
neighbor 2001:DB8:ACAD:1::2 activate
neighbor 2001:DB8:ACAD:C::1 activate
exit-address-family
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
```

```
ipv6 router ospf 10
router-id 3.3.3.3
redistribute bgp 3
control-plane
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Router4:

```
Current configuration : 2663 bytes
Last configuration change at 22:20:23 UTC Thu Nov 14 2024
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
hostname R4
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
ipv6 unicast-routing
subscriber templating
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FDO211216BL
spanning-tree extend system-id
redundancy
mode none
vlan internal allocation policy ascending
interface Loopback0
ip address 10.0.0.2 255.255.255.255
ipv6 address FE80::2 link-local
ipv6 address 2001:DB8:ACAD:1::2/128
ipv6 enable
ipv6 ospf 10 area 0
ipv6 ospf network point-to-point
interface GigabitEthernet0/0/0
ip address 192.168.4.1 255.255.255.252
negotiation auto
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:ACAD:E::1/64
```

```
ipv6 enable
ipv6 ospf 10 area 1
interface GigabitEthernet0/0/1
ip address 192.168.7.2 255.255.255.252
negotiation auto
ipv6 address FE80::2 link-local
ipv6 address 2001:DB8:ACAD:A2::2/64
ipv6 enable
ipv6 ospf 10 area 1
interface Serial0/1/0
no ip address
interface Serial0/1/1
no ip address
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto
interface Vlan1
no ip address
router ospf 10
router-id 4.4.4.4
redistribute bgp 3 metric 10 subnets
network 10.0.0.2 0.0.0.0 area 0
network 192.168.4.0 0.0.0.3 area 1
network 192.168.7.0 0.0.0.3 area 1
router bgp 3
bgp router-id 4.4.4.4
bgp log-neighbor-changes
neighbor 10.0.0.1 remote-as 3
neighbor 10.0.0.1 update-source Loopback0
neighbor 2001:DB8:ACAD:1::1 remote-as 3
neighbor 2001:DB8:ACAD:1::1 update-source Loopback0
neighbor 2001:DB8:ACAD:E::2 remote-as 5
neighbor 192.168.4.2 remote-as 5
address-family ipv4
redistribute connected
redistribute ospf 10 metric 10
neighbor 10.0.0.1 activate
no neighbor 2001:DB8:ACAD:1::1 activate
no neighbor 2001:DB8:ACAD:E::2 activate
neighbor 192.168.4.2 activate
exit-address-family
address-family ipv6
redistribute connected
redistribute ospf 10 metric 10
neighbor 2001:DB8:ACAD:1::1 activate
neighbor 2001:DB8:ACAD:E::2 activate
```

```
exit-address-family
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 10
router-id 4.4.4.4
redistribute bgp 3 metric 10
control-plane
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Router5:

```
Current configuration : 2228 bytes
Last configuration change at 21:59:09 UTC Thu Nov 14 2024
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
hostname R5
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
ipv6 unicast-routing
subscriber templating
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FDO214421CF
spanning-tree extend system-id
redundancy
mode none
vlan internal allocation policy ascending
interface GigabitEthernet0/0/0
ip address 192.168.5.1 255.255.255.252
negotiation auto
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:ACAD:F::1/64
ipv6 enable
ipv6 eigrp 10
```

```
interface GigabitEthernet0/0/1
ip address 192.168.4.2 255.255.255.252
negotiation auto
ipv6 address FE80::2 link-local
ipv6 address 2001:DB8:ACAD:E::2/64
ipv6 enable
ipv6 eigrp 10
interface Serial0/1/0
no ip address
interface Serial0/1/1
no ip address
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto
interface Vlan1
no ip address
router eigrp 10
network 192.168.4.0 0.0.0.3
network 192.168.5.0 0.0.0.3
redistribute connected metric 10 0 255 255 1
redistribute bgp 5 metric 10 0 255 255 1
eigrp router-id 5.5.5.5
router bgp 5
bgp router-id 5.5.5.5
bgp log-neighbor-changes
neighbor 2001:DB8:ACAD:E::1 remote-as 3
neighbor 192.168.4.1 remote-as 3
address-family ipv4
redistribute connected metric 10
redistribute eigrp 10 metric 10
no neighbor 2001:DB8:ACAD:E::1 activate
neighbor 192.168.4.1 activate
exit-address-family
address-family ipv6
redistribute connected metric 10
redistribute eigrp 10 metric 10
neighbor 2001:DB8:ACAD:E::1 activate
exit-address-family
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router eigrp 10
eigrp router-id 5.5.5.5
redistribute bgp 5 metric 10 0 255 255 1
control-plane
```

```
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Router6:

```
Current configuration : 4014 bytes
Last configuration change at 23:02:07 UTC Thu Nov 14 2024
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname R6
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FDO21441WBL
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 192.168.6.1 255.255.255.0
negotiation auto
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:ACAD:A1::1/64
ipv6 enable
ipv6 eigrp 10
interface GigabitEthernet0/0/1
ip address 192.168.5.2 255.255.255.252
negotiation auto
ipv6 address FE80::2 link-local
ipv6 address 2001:DB8:ACAD:F::2/64
```

```
ipv6 enable
ipv6 eigrp 10
interface Serial0/1/0
interface Serial0/1/1
interface GigabitEthernet0/2/0
no ip address
shutdown
negotiation auto
interface GigabitEthernet0/2/1
no ip address
shutdown
negotiation auto
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router eigrp 10
network 192.168.5.0
network 192.168.6.0
eigrp router-id 6.6.6.6
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router eigrp 10
eigrp router-id 6.6.6.6
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Router7:

```
Current configuration : 3871 bytes
Last configuration change at 22:15:22 UTC Thu Nov 14 2024
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname R7
boot-start-marker
```

```
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FDO214421BX
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 192.168.7.1 255.255.255.252
negotiation auto
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:ACAD:A2::1/64
ipv6 enable
ipv6 ospf 10 area 1
interface GigabitEthernet0/0/1
ip address 192.168.3.2 255.255.255.252
negotiation auto
ipv6 address FE80::2 link-local
ipv6 address 2001:DB8:ACAD:D::2/64
ipv6 enable
ipv6 ospf 10 area 1
interface Serial0/1/0
interface Serial0/1/1
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router ospf 10
router-id 7.7.7.7
network 192.168.3.0 0.0.0.3 area 1
network 192.168.7.0 0.0.0.3 area 1
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
```

```
ipv6 router ospf 10
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Screenshots:

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/32 is subnetted, 2 subnets
D EX     10.0.0.1
           [170/256000256] via 192.168.1.2, 00:24:54, GigabitEthernet0/0/0
D EX     10.0.0.2
           [170/256000256] via 192.168.1.2, 00:24:54, GigabitEthernet0/0/0
      192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.0.0/24 is directly connected, GigabitEthernet0/0/1
L        192.168.0.2/32 is directly connected, GigabitEthernet0/0/1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/30 is directly connected, GigabitEthernet0/0/0
L        192.168.1.1/32 is directly connected, GigabitEthernet0/0/0
      192.168.2.0/30 is subnetted, 1 subnets
D        192.168.2.0 [90/3072] via 192.168.1.2, 00:25:47, GigabitEthernet0/0/0
      192.168.3.0/30 is subnetted, 1 subnets
D EX     192.168.3.0
           [170/256000256] via 192.168.1.2, 00:24:54, GigabitEthernet0/0/0
      192.168.4.0/30 is subnetted, 1 subnets
D EX     192.168.4.0
           [170/256000256] via 192.168.1.2, 00:24:54, GigabitEthernet0/0/0
      192.168.5.0/30 is subnetted, 1 subnets
D EX     192.168.5.0
           [170/256000256] via 192.168.1.2, 00:24:54, GigabitEthernet0/0/0
D EX  192.168.6.0/24
           [170/256000256] via 192.168.1.2, 00:24:54, GigabitEthernet0/0/0
      192.168.7.0/30 is subnetted, 1 subnets
D EX     192.168.7.0
           [170/256000256] via 192.168.1.2, 00:24:54, GigabitEthernet0/0/0
R1#show ipv6 route
IPv6 Routing Table - default - 13 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
EX  2001:DB8:ACAD:1::1/128 [170/256000256]
     via FE80::2, GigabitEthernet0/0/0
EX  2001:DB8:ACAD:1::2/128 [170/256000256]
     via FE80::2, GigabitEthernet0/0/0
C   2001:DB8:ACAD:A::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:DB8:ACAD:A::2/128 [0/0]
     via GigabitEthernet0/0/1, receive
C   2001:DB8:ACAD:B::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:B::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
D   2001:DB8:ACAD:C::/64 [90/3072]
     via FE80::2, GigabitEthernet0/0/0
EX  2001:DB8:ACAD:D::/64 [170/256000256]
     via FE80::2, GigabitEthernet0/0/0
EX  2001:DB8:ACAD:E::/64 [170/256000256]
     via FE80::2, GigabitEthernet0/0/0
EX  2001:DB8:ACAD:F::/64 [170/256000256]
     via FE80::2, GigabitEthernet0/0/0
EX  2001:DB8:ACAD:A1::/64 [170/256000256]
     via FE80::2, GigabitEthernet0/0/0
EX  2001:DB8:ACAD:A2::/64 [170/256000256]
     via FE80::2, GigabitEthernet0/0/0
L   FF00::/8 [0/0]
     via Null0, receive
R1#
```

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/32 is subnetted, 2 subnets
B        10.0.0.1 [20/0] via 192.168.2.2, 00:26:20
B        10.0.0.2 [20/10] via 192.168.2.2, 00:26:20
D     192.168.0.0/24 [90/3072] via 192.168.1.1, 00:47:53, GigabitEthernet0/0/1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/30 is directly connected, GigabitEthernet0/0/1
L        192.168.1.2/32 is directly connected, GigabitEthernet0/0/1
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/30 is directly connected, GigabitEthernet0/0/0
L        192.168.2.1/32 is directly connected, GigabitEthernet0/0/0
      192.168.3.0/30 is subnetted, 1 subnets
B        192.168.3.0 [20/0] via 192.168.2.2, 00:26:20
      192.168.4.0/30 is subnetted, 1 subnets
B        192.168.4.0 [20/10] via 192.168.2.2, 00:26:20
      192.168.5.0/30 is subnetted, 1 subnets
B        192.168.5.0 [20/0] via 192.168.2.2, 00:26:20
B     192.168.6.0/24 [20/0] via 192.168.2.2, 00:26:20
      192.168.7.0/30 is subnetted, 1 subnets
B        192.168.7.0 [20/10] via 192.168.2.2, 00:26:20
R2#show ipv6 route
IPv6 Routing Table - default - 13 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
B   2001:DB8:ACAD:1::1/128 [20/0]
     via FE80::1, GigabitEthernet0/0/0
B   2001:DB8:ACAD:1::2/128 [20/10]
     via FE80::1, GigabitEthernet0/0/0
D   2001:DB8:ACAD:A::/64 [90/3072]
     via FE80::1, GigabitEthernet0/0/1
C   2001:DB8:ACAD:B::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:DB8:ACAD:B::2/128 [0/0]
     via GigabitEthernet0/0/1, receive
C   2001:DB8:ACAD:C::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:C::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
B   2001:DB8:ACAD:D::/64 [20/0]
     via FE80::1, GigabitEthernet0/0/0
B   2001:DB8:ACAD:E::/64 [20/10]
     via FE80::1, GigabitEthernet0/0/0
B   2001:DB8:ACAD:F::/64 [20/0]
     via FE80::1, GigabitEthernet0/0/0
B   2001:DB8:ACAD:A1::/64 [20/0]
     via FE80::1, GigabitEthernet0/0/0
B   2001:DB8:ACAD:A2::/64 [20/10]
     via FE80::1, GigabitEthernet0/0/0
L   FF00::/8 [0/0]
     via Null0, receive
R2#
```

```
C:\Users\Alex Huhdanpaa>tracert 2001:db8:acad:a1::2

Tracing route to 2001:db8:acad:a1::2 over a maximum of 30 hops

  1    <1 ms    <1 ms    <1 ms  2001:db8:acad:a::2
  2     1 ms    <1 ms    <1 ms  2001:db8:acad:b::2
  3     1 ms    <1 ms    <1 ms  2001:db8:acad:c::2
  4     1 ms    <1 ms     1 ms  2001:db8:acad:d::2
  5     1 ms    <1 ms    <1 ms  2001:db8:acad:a2::2
  6     1 ms    <1 ms    <1 ms  2001:db8:acad:e::2
  7     1 ms     1 ms     1 ms  2001:db8:acad:f::2
  8     4 ms     1 ms     1 ms  2001:db8:acad:a1::2

Trace complete.

C:\Users\Alex Huhdanpaa>tracert 192.168.6.2

Tracing route to DESKTOP-JT6LALA [192.168.6.2]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  192.168.0.2
  2    <1 ms    <1 ms    <1 ms  192.168.1.2
  3    <1 ms    <1 ms    <1 ms  192.168.2.2
  4     1 ms    <1 ms     1 ms  192.168.3.2
  5    <1 ms    <1 ms    <1 ms  192.168.7.2
  6    <1 ms    <1 ms    <1 ms  192.168.4.2
  7     1 ms     1 ms     1 ms  192.168.5.2
  8     1 ms     1 ms     1 ms  DESKTOP-JT6LALA [192.168.6.2]

Trace complete.
```

```
C:\Users\Brandon>tracert 192.168.0.1

Tracing route to DESKTOP-RDJTFSD [192.168.0.1]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  192.168.6.1
  2     1 ms    <1 ms    <1 ms  192.168.5.1
  3     1 ms    <1 ms    <1 ms  192.168.4.1
  4     1 ms    <1 ms    <1 ms  192.168.7.1
  5    <1 ms    <1 ms    <1 ms  192.168.3.1
  6     1 ms     1 ms     1 ms  192.168.2.1
  7     1 ms     1 ms     1 ms  192.168.1.1
  8     1 ms     1 ms     1 ms  DESKTOP-RDJTFSD [192.168.0.1]

Trace complete.

C:\Users\Brandon>tracert 2001:db8:acad:a::1

Tracing route to 2001:db8:acad:a::1 over a maximum of 30 hops

  1     1 ms    <1 ms    <1 ms  2001:db8:acad:a1::1
  2     1 ms    <1 ms    <1 ms  2001:db8:acad:f::1
  3    <1 ms    <1 ms    <1 ms  2001:db8:acad:e::1
  4     1 ms     1 ms    <1 ms  2001:db8:acad:a2::1
  5     1 ms    <1 ms    <1 ms  2001:db8:acad:d::1
  6     1 ms     1 ms     1 ms  2001:db8:acad:c::1
  7     1 ms     1 ms     1 ms  2001:db8:acad:b::1
  8     1 ms     1 ms     1 ms  2001:db8:acad:a::1

Trace complete.
```

```
BGP neighbor is 10.0.0.2,  remote AS 3, internal link
BGP neighbor is 192.168.2.1,  remote AS 2, external link
BGP neighbor is 2001:DB8:ACAD:1::2,  remote AS 3, internal link
BGP neighbor is 2001:DB8:ACAD:C::1,  remote AS 2, external link
BGP neighbor is 10.0.0.1,  remote AS 3, internal link
BGP neighbor is 192.168.4.2,  remote AS 5, external link
BGP neighbor is 2001:DB8:ACAD:1::1,  remote AS 3, internal link
BGP neighbor is 2001:DB8:ACAD:E::2,  remote AS 5, external link
```

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/32 is subnetted, 2 subnets
C        10.0.0.1 is directly connected, Loopback0
O IA     10.0.0.2 [110/3] via 192.168.3.2, 00:30:32, GigabitEthernet0/0/0
B     192.168.0.0/24 [20/10] via 192.168.2.1, 00:30:21
      192.168.1.0/30 is subnetted, 1 subnets
B        192.168.1.0 [20/10] via 192.168.2.1, 00:30:21
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/30 is directly connected, GigabitEthernet0/0/1
L        192.168.2.2/32 is directly connected, GigabitEthernet0/0/1
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.0/30 is directly connected, GigabitEthernet0/0/0
L        192.168.3.1/32 is directly connected, GigabitEthernet0/0/0
      192.168.4.0/30 is subnetted, 1 subnets
O        192.168.4.0 [110/3] via 192.168.3.2, 00:30:32, GigabitEthernet0/0/0
      192.168.5.0/30 is subnetted, 1 subnets
O E2     192.168.5.0 [110/10] via 192.168.3.2, 00:30:32, GigabitEthernet0/0/0
O E2  192.168.6.0/24 [110/10] via 192.168.3.2, 00:30:32, GigabitEthernet0/0/0
      192.168.7.0/30 is subnetted, 1 subnets
O        192.168.7.0 [110/2] via 192.168.3.2, 00:30:32, GigabitEthernet0/0/0
R3#show ipv6 route
IPv6 Routing Table - default - 13 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
LC  2001:DB8:ACAD:1::1/128 [0/0]
     via Loopback0, receive
OI  2001:DB8:ACAD:1::2/128 [110/2]
     via FE80::2, GigabitEthernet0/0/0
B   2001:DB8:ACAD:A::/64 [20/10]
     via FE80::2, GigabitEthernet0/0/1
B   2001:DB8:ACAD:B::/64 [20/10]
     via FE80::2, GigabitEthernet0/0/1
C   2001:DB8:ACAD:C::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:DB8:ACAD:C::2/128 [0/0]
     via GigabitEthernet0/0/1, receive
C   2001:DB8:ACAD:D::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:D::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
O   2001:DB8:ACAD:E::/64 [110/3]
     via FE80::2, GigabitEthernet0/0/0
OE2 2001:DB8:ACAD:F::/64 [110/10]
     via FE80::2, GigabitEthernet0/0/0
OE2 2001:DB8:ACAD:A1::/64 [110/10]
     via FE80::2, GigabitEthernet0/0/0
O   2001:DB8:ACAD:A2::/64 [110/2]
     via FE80::2, GigabitEthernet0/0/0
L   FF00::/8 [0/0]
     via Null0, receive
R3#
```

```
R4#show bgp
BGP table version is 36, local router ID is 4.4.4.4
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 * i 10.0.0.1/32      10.0.0.1                 0    100      0 ?
 *>                   192.168.7.1             10         32768 ?
 * i 10.0.0.2/32      192.168.3.2             10    100      0 ?
 *>                   0.0.0.0                  0         32768 ?
 r>i 192.168.0.0      192.168.2.1             10    100      0 2 ?
 r>i 192.168.1.0/30   192.168.2.1             10    100      0 2 ?
 * i 192.168.2.0/30   10.0.0.1                 0    100      0 ?
 *>                   192.168.7.1             10         32768 ?
 * i 192.168.3.0/30   10.0.0.1                 0    100      0 ?
 *>                   192.168.7.1             10         32768 ?
 * i 192.168.4.0/30   192.168.3.2             10    100      0 ?
 *                    192.168.4.2             10              0 5 ?
 *>                   0.0.0.0                  0         32768 ?
 *>  192.168.5.0/30   192.168.4.2             10              0 5 ?
     Network          Next Hop            Metric LocPrf Weight Path
 *>  192.168.6.0      192.168.4.2             10              0 5 ?
 * i 192.168.7.0/30   192.168.3.2             10    100      0 ?
 *>                   0.0.0.0                  0         32768 ?
R4#show bgp ipv6 unicast
BGP table version is 63, local router ID is 4.4.4.4
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 * i 2001:DB8:ACAD:1::1/128
                      2001:DB8:ACAD:1::1
                                               0    100      0 ?
 *>                   FE80::1                 10         32768 ?
 * i 2001:DB8:ACAD:1::2/128
                      2001:DB8:ACAD:1::1
                                              10    100      0 ?
 *>                   ::                       0         32768 ?
 r>i 2001:DB8:ACAD:A::/64
                      2001:DB8:ACAD:C::1
                                              10    100      0 2 ?
 r>i 2001:DB8:ACAD:B::/64
                      2001:DB8:ACAD:C::1
                                              10    100      0 2 ?
     Network          Next Hop            Metric LocPrf Weight Path
 * i 2001:DB8:ACAD:C::/64
                      2001:DB8:ACAD:1::1
                                               0    100      0 ?
 *>                   FE80::1                 10         32768 ?
 * i 2001:DB8:ACAD:D::/64
                      2001:DB8:ACAD:1::1
                                               0    100      0 ?
 *>                   FE80::1                 10         32768 ?
 * i 2001:DB8:ACAD:E::/64
                      2001:DB8:ACAD:1::1
                                              10    100      0 ?
 *                    2001:DB8:ACAD:E::2
                                              10              0 5 ?
 *>                   ::                       0         32768 ?
 *>  2001:DB8:ACAD:F::/64
                      2001:DB8:ACAD:E::2
                                              10              0 5 ?
 *>  2001:DB8:ACAD:A1::/64
                      2001:DB8:ACAD:E::2
                                              10              0 5 ?
 * i 2001:DB8:ACAD:A2::/64
                      2001:DB8:ACAD:1::1
     Network          Next Hop            Metric LocPrf Weight Path
                                              10    100      0 ?
 *>                   ::                       0         32768 ?
R4#
```

```
R5#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/32 is subnetted, 2 subnets
B        10.0.0.1 [20/10] via 192.168.4.1, 00:37:51
B        10.0.0.2 [20/0] via 192.168.4.1, 01:00:04
B     192.168.0.0/24 [20/0] via 192.168.4.1, 00:37:21
      192.168.1.0/30 is subnetted, 1 subnets
B        192.168.1.0 [20/0] via 192.168.4.1, 00:37:21
      192.168.2.0/30 is subnetted, 1 subnets
B        192.168.2.0 [20/10] via 192.168.4.1, 00:37:51
      192.168.3.0/30 is subnetted, 1 subnets
B        192.168.3.0 [20/10] via 192.168.4.1, 00:38:24
      192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.4.0/30 is directly connected, GigabitEthernet0/0/1
L        192.168.4.2/32 is directly connected, GigabitEthernet0/0/1
      192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.5.0/30 is directly connected, GigabitEthernet0/0/0
L        192.168.5.1/32 is directly connected, GigabitEthernet0/0/0
D     192.168.6.0/24 [90/3072] via 192.168.5.2, 01:01:53, GigabitEthernet0/0/0
      192.168.7.0/30 is subnetted, 1 subnets
B        192.168.7.0 [20/0] via 192.168.4.1, 01:00:04
R5#show ipv6 route
IPv6 Routing Table - default - 13 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
B   2001:DB8:ACAD:1::1/128 [20/10]
     via FE80::1, GigabitEthernet0/0/1
B   2001:DB8:ACAD:1::2/128 [20/0]
     via FE80::1, GigabitEthernet0/0/1
B   2001:DB8:ACAD:A::/64 [20/0]
     via FE80::1, GigabitEthernet0/0/1
B   2001:DB8:ACAD:B::/64 [20/0]
     via FE80::1, GigabitEthernet0/0/1
B   2001:DB8:ACAD:C::/64 [20/10]
     via FE80::1, GigabitEthernet0/0/1
B   2001:DB8:ACAD:D::/64 [20/10]
     via FE80::1, GigabitEthernet0/0/1
C   2001:DB8:ACAD:E::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:DB8:ACAD:E::2/128 [0/0]
     via GigabitEthernet0/0/1, receive
C   2001:DB8:ACAD:F::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:F::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
D   2001:DB8:ACAD:A1::/64 [90/3072]
     via FE80::2, GigabitEthernet0/0/0
B   2001:DB8:ACAD:A2::/64 [20/0]
     via FE80::1, GigabitEthernet0/0/1
L   FF00::/8 [0/0]
     via Null0, receive
R5#
```

```
R6#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/32 is subnetted, 2 subnets
D EX     10.0.0.1
            [170/256000256] via 192.168.5.1, 00:38:47, GigabitEthernet0/0/1
D EX     10.0.0.2
            [170/256000256] via 192.168.5.1, 01:01:00, GigabitEthernet0/0/1
D EX  192.168.0.0/24
            [170/256000256] via 192.168.5.1, 00:38:17, GigabitEthernet0/0/1
      192.168.1.0/30 is subnetted, 1 subnets
D EX     192.168.1.0
            [170/256000256] via 192.168.5.1, 00:38:17, GigabitEthernet0/0/1
      192.168.2.0/30 is subnetted, 1 subnets
D EX     192.168.2.0
            [170/256000256] via 192.168.5.1, 00:38:47, GigabitEthernet0/0/1
      192.168.3.0/30 is subnetted, 1 subnets
D EX     192.168.3.0
            [170/256000256] via 192.168.5.1, 00:39:19, GigabitEthernet0/0/1
      192.168.4.0/30 is subnetted, 1 subnets
D        192.168.4.0 [90/3072] via 192.168.5.1, 01:02:51, GigabitEthernet0/0/1
      192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.5.0/30 is directly connected, GigabitEthernet0/0/1
L        192.168.5.2/32 is directly connected, GigabitEthernet0/0/1
      192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.6.0/24 is directly connected, GigabitEthernet0/0/0
L        192.168.6.1/32 is directly connected, GigabitEthernet0/0/0
      192.168.7.0/30 is subnetted, 1 subnets
D EX     192.168.7.0
            [170/256000256] via 192.168.5.1, 01:01:00, GigabitEthernet0/0/1
R6#show ipv6 route
IPv6 Routing Table - default - 13 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
EX  2001:DB8:ACAD:1::1/128 [170/256000256]
     via FE80::1, GigabitEthernet0/0/1
EX  2001:DB8:ACAD:1::2/128 [170/256000256]
     via FE80::1, GigabitEthernet0/0/1
EX  2001:DB8:ACAD:A::/64 [170/256000256]
     via FE80::1, GigabitEthernet0/0/1
EX  2001:DB8:ACAD:B::/64 [170/256000256]
     via FE80::1, GigabitEthernet0/0/1
EX  2001:DB8:ACAD:C::/64 [170/256000256]
     via FE80::1, GigabitEthernet0/0/1
EX  2001:DB8:ACAD:D::/64 [170/256000256]
     via FE80::1, GigabitEthernet0/0/1
D   2001:DB8:ACAD:E::/64 [90/3072]
     via FE80::1, GigabitEthernet0/0/1
C   2001:DB8:ACAD:F::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:DB8:ACAD:F::2/128 [0/0]
     via GigabitEthernet0/0/1, receive
C   2001:DB8:ACAD:A1::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:A1::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
EX  2001:DB8:ACAD:A2::/64 [170/256000256]
     via FE80::1, GigabitEthernet0/0/1
L   FF00::/8 [0/0]
     via Null0, receive
R6#
```

```
R3#show bgp
BGP table version is 11, local router ID is 3.3.3.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 * i 10.0.0.1/32      192.168.7.1              10    100      0 ?
 *>                   0.0.0.0                   0         32768 ?
 * i 10.0.0.2/32      10.0.0.2                  0    100      0 ?
 *>                   192.168.3.2              10         32768 ?
 *>  192.168.0.0      192.168.2.1              10              0 2 ?
 *>  192.168.1.0/30   192.168.2.1              10              0 2 ?
 * i 192.168.2.0/30   192.168.7.1              10    100      0 ?
 *                    192.168.2.1              10              0 2 ?
 *>                   0.0.0.0                   0         32768 ?
 * i 192.168.3.0/30   192.168.7.1              10    100      0 ?
 *>                   0.0.0.0                   0         32768 ?
 * i 192.168.4.0/30   10.0.0.2                  0    100      0 ?
 *>                   192.168.3.2              10         32768 ?
 r>i 192.168.5.0/30   192.168.4.2              10    100      0 5 ?
     Network          Next Hop            Metric LocPrf Weight Path
 r>i 192.168.6.0      192.168.4.2              10    100      0 5 ?
 * i 192.168.7.0/30   10.0.0.2                  0    100      0 ?
 *>                   192.168.3.2              10         32768 ?
R3#show bgp ipv6 unicast
BGP table version is 36, local router ID is 3.3.3.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 * i 2001:DB8:ACAD:1::1/128
                      2001:DB8:ACAD:1::2
                                           10    100      0 ?
 *>                   ::                    0         32768 ?
 * i 2001:DB8:ACAD:1::2/128
                      2001:DB8:ACAD:1::2
                                            0    100      0 ?
 *>                   FE80::2              10         32768 ?
 *>  2001:DB8:ACAD:A::/64
                      2001:DB8:ACAD:C::1
                                           10              0 2 ?
 *>  2001:DB8:ACAD:B::/64
                      2001:DB8:ACAD:C::1
                                           10              0 2 ?
     Network          Next Hop            Metric LocPrf Weight Path
 * i 2001:DB8:ACAD:C::/64
                      2001:DB8:ACAD:1::2
                                           10    100      0 ?
 *                    2001:DB8:ACAD:C::1
                                           10              0 2 ?
 *>                   ::                    0         32768 ?
 * i 2001:DB8:ACAD:D::/64
                      2001:DB8:ACAD:1::2
                                           10    100      0 ?
 *>                   ::                    0         32768 ?
 * i 2001:DB8:ACAD:E::/64
                      2001:DB8:ACAD:1::2
                                            0    100      0 ?
 *>                   FE80::2              10         32768 ?
 r>i 2001:DB8:ACAD:F::/64
                      2001:DB8:ACAD:E::2
                                           10    100      0 5 ?
 r>i 2001:DB8:ACAD:A1::/64
                      2001:DB8:ACAD:E::2
                                           10    100      0 5 ?
 * i 2001:DB8:ACAD:A2::/64
                      2001:DB8:ACAD:1::2
     Network          Next Hop            Metric LocPrf Weight Path
                                            0    100      0 ?
 *>                   FE80::2              10         32768 ?
R3#
```

```
R4#show bgp
BGP table version is 36, local router ID is 4.4.4.4
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 * i 10.0.0.1/32      10.0.0.1                  0    100      0 ?
 *>                   192.168.7.1              10         32768 ?
 * i 10.0.0.2/32      192.168.3.2              10    100      0 ?
 *>                   0.0.0.0                   0         32768 ?
 r>i 192.168.0.0      192.168.2.1              10    100      0 2 ?
 r>i 192.168.1.0/30   192.168.2.1              10    100      0 2 ?
 * i 192.168.2.0/30   10.0.0.1                  0    100      0 ?
 *>                   192.168.7.1              10         32768 ?
 * i 192.168.3.0/30   10.0.0.1                  0    100      0 ?
 *>                   192.168.7.1              10         32768 ?
 * i 192.168.4.0/30   192.168.3.2              10    100      0 ?
 *                    192.168.4.2              10              0 5 ?
 *>                   0.0.0.0                   0         32768 ?
     Network          Next Hop            Metric LocPrf Weight Path
 *>  192.168.5.0/30   192.168.4.2              10              0 5 ?
     Network          Next Hop            Metric LocPrf Weight Path
 *>  192.168.6.0      192.168.4.2              10              0 5 ?
 * i 192.168.7.0/30   192.168.3.2              10    100      0 ?
 *>                   0.0.0.0                   0         32768 ?
R4#show bgp ipv6 unicast
BGP table version is 63, local router ID is 4.4.4.4
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 * i 2001:DB8:ACAD:1::1/128
                      2001:DB8:ACAD:1::1
                                            0    100      0 ?
 *>                   FE80::1              10         32768 ?
 * i 2001:DB8:ACAD:1::2/128
                      2001:DB8:ACAD:1::1
                                           10    100      0 ?
 *>                   ::                    0         32768 ?
 r>i 2001:DB8:ACAD:A::/64
                      2001:DB8:ACAD:C::1
                                           10    100      0 2 ?
 r>i 2001:DB8:ACAD:B::/64
                      2001:DB8:ACAD:C::1
                                           10    100      0 2 ?
     Network          Next Hop            Metric LocPrf Weight Path
 * i 2001:DB8:ACAD:C::/64
                      2001:DB8:ACAD:1::1
                                            0    100      0 ?
 *>                   FE80::1              10         32768 ?
 * i 2001:DB8:ACAD:D::/64
                      2001:DB8:ACAD:1::1
                                            0    100      0 ?
 *>                   FE80::1              10         32768 ?
 * i 2001:DB8:ACAD:E::/64
                      2001:DB8:ACAD:1::1
                                           10    100      0 ?
 *                    2001:DB8:ACAD:E::2
                                           10              0 5 ?
 *>                   ::                    0         32768 ?
 *>  2001:DB8:ACAD:F::/64
                      2001:DB8:ACAD:E::2
                                           10              0 5 ?
 *>  2001:DB8:ACAD:A1::/64
                      2001:DB8:ACAD:E::2
                                           10              0 5 ?
 * i 2001:DB8:ACAD:A2::/64
                      2001:DB8:ACAD:1::1
     Network          Next Hop            Metric LocPrf Weight Path
                                           10    100      0 ?
 *>                   ::                    0         32768 ?
R4#
```

```
R7#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/32 is subnetted, 2 subnets
O IA     10.0.0.1 [110/2] via 192.168.3.1, 00:40:45, GigabitEthernet0/0/1
O IA     10.0.0.2 [110/2] via 192.168.7.2, 01:04:02, GigabitEthernet0/0/0
O E2  192.168.0.0/24 [110/10] via 192.168.3.1, 00:40:35, GigabitEthernet0/0/1
      192.168.1.0/30 is subnetted, 1 subnets
O E2     192.168.1.0 [110/10] via 192.168.3.1, 00:40:35, GigabitEthernet0/0/1
      192.168.2.0/30 is subnetted, 1 subnets
O        192.168.2.0 [110/2] via 192.168.3.1, 00:40:45, GigabitEthernet0/0/1
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.0/30 is directly connected, GigabitEthernet0/0/1
L        192.168.3.2/32 is directly connected, GigabitEthernet0/0/1
      192.168.4.0/30 is subnetted, 1 subnets
O        192.168.4.0 [110/2] via 192.168.7.2, 01:04:02, GigabitEthernet0/0/0
      192.168.5.0/30 is subnetted, 1 subnets
O E2     192.168.5.0 [110/10] via 192.168.7.2, 01:02:48, GigabitEthernet0/0/0
O E2  192.168.6.0/24 [110/10] via 192.168.7.2, 01:02:48, GigabitEthernet0/0/0
      192.168.7.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.7.0/30 is directly connected, GigabitEthernet0/0/0
L        192.168.7.1/32 is directly connected, GigabitEthernet0/0/0
R7#show ipv6 route
IPv6 Routing Table - default - 13 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
OI  2001:DB8:ACAD:1::1/128 [110/1]
     via FE80::1, GigabitEthernet0/0/1
OI  2001:DB8:ACAD:1::2/128 [110/1]
     via FE80::2, GigabitEthernet0/0/0
OE2 2001:DB8:ACAD:A::/64 [110/1]
     via FE80::1, GigabitEthernet0/0/1
OE2 2001:DB8:ACAD:B::/64 [110/1]
     via FE80::1, GigabitEthernet0/0/1
O   2001:DB8:ACAD:C::/64 [110/2]
     via FE80::1, GigabitEthernet0/0/1
C   2001:DB8:ACAD:D::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:DB8:ACAD:D::2/128 [0/0]
     via GigabitEthernet0/0/1, receive
O   2001:DB8:ACAD:E::/64 [110/2]
     via FE80::2, GigabitEthernet0/0/0
OE2 2001:DB8:ACAD:F::/64 [110/10]
     via FE80::2, GigabitEthernet0/0/0
OE2 2001:DB8:ACAD:A1::/64 [110/10]
     via FE80::2, GigabitEthernet0/0/0
C   2001:DB8:ACAD:A2::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:A2::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
L   FF00::/8 [0/0]
     via Null0, receive
R7#
```

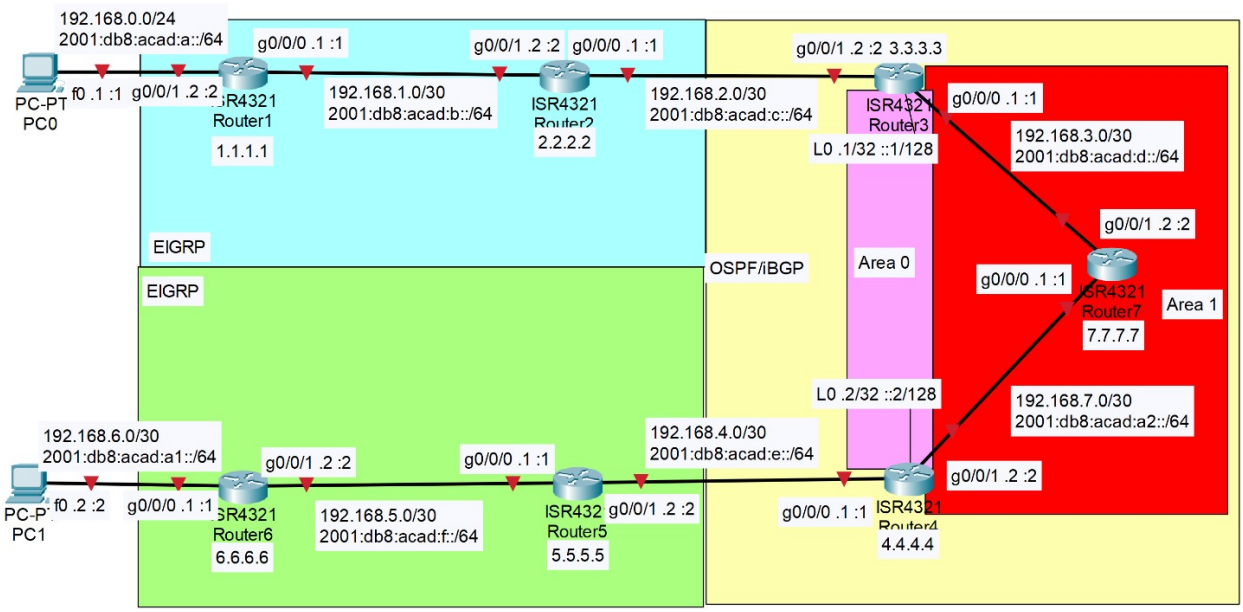
Problems:

I ran in to a number of issues, among them were improperly configured OSPF network commands for the point to point virtual connection between routers 3 and 4, which caused the iBGP routers to not able to peer, this was caused by the network commands pointing towards the other loopback interface, and not their own, and was fixed by using a no network command then reconfiguring the network commands. Another issue was unreliability in pinging, where there were two conflicting routes to the same place, one from the iBGP peer, and one from OSPF external 2 routes that were redistributed by BGP, and then being redistributed back in. this was resolved by removing the command that redistributed commands from OSPF back into BGP. The final issue. The final issue was in configurations copied into the router I had accidentally written a network prefix onto the end of a address-family ipv4 neighbor command which made it not build the adjacency this was resolved by removing the network prefix from the command

Conclusion:

This lab was a major success, thanks to previous configurations of all Routing Protocols except for iBGP which has very similar configuration eBGP which as said I have already configured, this meant that with the exception of the minor hiccups seen in the problems section the configuration went smoothly, and hopefully demonstrates an aptitude for modern corporate networking.

Diagram and Addressing:



PC1	----	F0	2001:db8:acad:a::1/64	192.168.0.1	255.255.255.0
R1	----	G0/0/1	2001:db8:acad:a::2/64	192.168.0.2	255.255.255.0
R1	EIGRP	G0/0/0	2001:db8:acad:b::1/64	192.168.1.1	255.255.255.252
R2	EIGRP	G0/0/1	2001:db8:acad:b::2/64	192.168.1.2	255.255.255.252
R2	EIGRP/BGP	G0/0/0	2001:db8:acad:c::1/64	192.168.2.1	255.255.255.252
R3	OSPF/BGP	G0/0/1	2001:db8:acad:c::2/64	192.168.2.2	255.255.255.252
R3	OSPF	G0/0/0	2001:db8:acad:d::1/64	192.168.3.1	255.255.255.252
R4	OSPF	G0/0/1	2001:db8:acad:a2::2/64	192.168.7.2	255.255.255.252
R4	OSPF/BGP	G0/0/0	2001:db8:acad:e::1/64	192.168.4.1	255.255.255.252
R5	EIGRP/BGP	G0/0/1	2001:db8:acad:e::2/64	192.168.4.2	255.255.255.252
R5	EIGRP	G0/0/0	2001:db8:acad:f::1/64	192.168.5.1	255.255.255.252
R6	EIGRP	G0/0/1	2001:db8:acad:f::2/64	192.168.5.2	255.255.255.252
R6	----	G0/0/0	2001:db8:acad:a1::1/64	192.168.6.1	255.255.255.0
PC2	----	F0	2001:db8:acad:a1::2/64	192.168.6.2	255.255.255.0
R7	OSPF	G0/0/0	2001:db8:acad:d::2/64	192.168.3.2	255.255.255.252
R7	OSPF	G0/0/1	2001:db8:acad:a2::1/64	192.168.7.1	255.255.255.252

Signoff:

CCNP eBGP Physical Lab Stamp Off

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